

Task 1: Encrypt and Decrypt a Message Using AES

Objective

To encrypt the message MyZomatoPassword2024 using AES and verify that decryption recovers the original message.

Given Information

Parameter	Value
Plaintext	MyZomatoPassword2024
Provided key	cyber1234
Algorithm	AES
Tool	Devglan AES Encryption/Decryption

Procedure

1. Open the AES encryption tool: Devglan AES Encryption/Decryption
2. Enter the plaintext: MyZomatoPassword2024.
3. Enter the provided key: cyber1234.
4. Select the required AES mode and padding options supported by the tool.
5. Click the encryption button and record the ciphertext.
6. Enter the ciphertext into the decryption section.
7. Use the same key and compatible settings.
8. Decrypt the ciphertext and compare the result with the original message.

Observation Table

Operation	Value
Original message	MyZomatoPassword2024
Key supplied	cyber1234

Encryption	Ciphertext generated by the tool
Decryption	Recovered plaintext
Verification	Recovered plaintext should match the original

Task 2: Three Daily-Use Apps and Their Use of Encryption

Objective

To identify how commonly used applications protect user information through encryption.

Application	How encryption is used
WhatsApp	End-to-end encryption protects the contents of personal messages and calls between communicating users.
Paytm	HTTPS/TLS helps protect data exchanged between the app and its servers. Payment credentials and sensitive operations also rely on additional security controls.
Instagram	HTTPS/TLS protects data transmitted between the app and Instagram's servers.

Explanation

Encryption helps protect information against unauthorized interception. However, encryption does not prevent every type of attack. Account security, device protection, authentication, and secure server design are also important.

Task 3: Comparison of AES and RSA

Objective

To compare symmetric and asymmetric encryption based on key usage, speed, and applications.

Feature	AES	RSA
Type	Symmetric encryption	Asymmetric cryptography
Key usage	Same secret key for encryption and decryption	Public/private key pair
Speed	Generally fast for bulk data	Generally slower for large data
Key distribution	Secret key must be shared securely	Public key can be distributed openly
Typical use	File, disk and communication encryption	Key establishment in some schemes and digital signatures
Common key sizes	128, 192, 256 bits	Commonly 2048 bits or higher, depending on requirements

Explanation

AES is efficient for encrypting large amounts of data. RSA uses a public/private key pair and is computationally more expensive, so it is generally not used to encrypt large files directly.

Modern secure communication systems often use hybrid cryptography: asymmetric cryptography helps establish or protect key material, while symmetric encryption protects the actual data.

Task 4: Generate an RSA Key Pair and Encrypt a Message

Objective

To generate an RSA public/private key pair, encrypt BookMyShowTickets using the public key, and decrypt it using the private key.

Procedure

Step 1: Open the RSA tool.

- Devglan RSA Encryption/Decryption

Step 2: Generate the keys.

- Select the RSA key generation option.
- Generate a public key and private key pair.
- Copy both keys into a local text file for reference.

Step 3: Encrypt the message.

Enter:

- BookMyShowTickets

Use the generated public key to encrypt the message.

Step 4: Decrypt the ciphertext.

- Copy the generated ciphertext into the decryption section. Use the corresponding private key to decrypt it.

Step 5: Verify the result.

- Compare the decrypted text with the original message.

Observation Table

Parameter	Value
Algorithm	RSA
Original plaintext	BookMyShowTickets
Public key	Generated using the tool

Private key	Generated using the tool
Ciphertext	Copy the actual output
Decrypted text	BookMyShowTickets
Result	Successful if the original message is recovered

Security Note

For a classroom demonstration, an online tool can illustrate RSA encryption. Do not upload real passwords, private keys, or confidential information to an online encryption website. Use a local trusted tool for sensitive data.

Task 5: Why Asymmetric Cryptography Is Used in Secure Online Payments

AI-Generated Explanation

Asymmetric cryptography is useful in secure online payments because it allows two parties to establish trust without first sharing a secret encryption key directly. It uses a public key and a mathematically related private key. The public key can be shared, while the private key must remain protected.

In modern online payment systems, asymmetric cryptography is commonly used as part of secure communication protocols, such as TLS. These protocols authenticate servers and establish shared session keys. Symmetric encryption is then used to protect the payment data exchanged during the session because it is efficient for large amounts of information.

Digital signatures can also help verify the authenticity and integrity of certain messages or transactions. However, asymmetric encryption alone does not guarantee that a payment is safe. Secure authentication, fraud

detection, tokenization, and proper handling of payment information are also necessary.

Example from Daily App Usage

When I use Paytm to make an online payment, the app communicates with its servers over a secure HTTPS connection. TLS uses cryptographic mechanisms to authenticate the server and establish protected communication. Symmetric encryption then protects the data transmitted during the session.

This helps reduce the risk of someone intercepting and reading sensitive information while it is being transmitted.